A Technical Introduction to Quantum Computing

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Introduction

From the dawn of 'computing' by human beings, then Alan Turing's Enigma Machine, the concept of **'computing'** has evolved greatly. We now stand at the precipice of another step, quantum computing from conventional classical computing. Quantum Computing is treated as a black-box of the future technology for the layman. This research project aims to provide a more technical understanding and review the current state of the field. Foremost, *"Quantum Computing and Quantum information is an area of study based on the idea that information is ultimately a physical quantity, and that by using quantum mechanics we can process information in fundamentally new ways. [8]"*. Whereas a device capable of quantum computation can be defined as such,

"A quantum computer is a physical device that uses quantum mechanics to perform computations; it manipulates information stored in quantum bits(qubits) through the application of unitary transformations and measurements.[8]"

Laying the Groundwork

Quantum computing is at the nexus of Mathematics, Physics, Quantum Physics, Material Sciences, Electronics and Computer Science. So, for a technical observation, a bare minimum familiarity if not a solid understanding of the following subjects are considered necessary:

1. **Mathematics :** Complex Numbers, Linear Algebra (Vector, Matrix, Linear Transform etc.)
2. **Physics :** Waves, Wave Functions etc.
3. **Computer Science :** Boolean Algebra, Logic Gates, Information Theory etc.
4. **Quantum Mechanics :** HUP, Quantum States & Notation, Entanglement, Interference etc.[7]

Fundamental Concepts

In classical computation, The fundamental unit of information is called 'the bit'. All data and their operations can be reduced to patterns of bits. For a quantum computer, an analogous unit known as the Quantum Bit(Qubit) is used to represent information. [8]

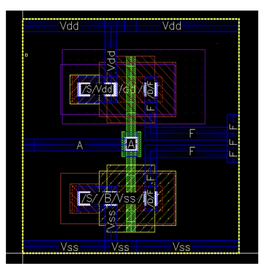
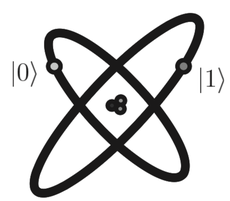
Layers of Abstraction	Classical Computer	Quantum Computer
Entity	2-State Number: b	2D Complex Vector: $\lvert\psi\rangle$
States / Bits	Classical Bits: | 1011 | Deterministic: Stored: 0 or 1 Measured: 0 or 1	Quantum Bits(Qubits): $\lvert 1011\rangle$ Non - Deterministic: Stored: Superposition of $\lvert 0\rangle$ and $\lvert 1\rangle$ Measured: $\lvert 0\rangle$ or $\lvert 1\rangle$
Storage	Small Charges, or, Magnetization	Quantum Property of a Particle E.g. Polarization, Spin etc.
Physical Layer		

Figure 1. CMOS Gate Circuit Layout in Semiconductor Layer in Cadence

Figure 2. Two Possible Electron States in a Quantum System[8]

Table 1. Conceptualizing Quantum Bits from Classical Bits [7]

How Quantum Computing Works

As stated, Qubits are the representations of quantum states. Quantum States are, *a two-dimensional vector that completely characterizes a quantum system.* $|\psi\rangle$ represents all possible states for a qubit between $|0\rangle$ to $|1\rangle$.

For a single qubit, we say mathematically:

$$|\psi\rangle = \alpha|\mathbf{0}\rangle + \beta|\mathbf{1}\rangle \equiv \begin{bmatrix}\alpha\\ \beta\end{bmatrix}$$

$$\text{Or, } |\psi\rangle = \cos\tfrac{\theta}{2}|\mathbf{0}\rangle + \mathbf{e}^{\mathbf{i}\phi}\sin\tfrac{\theta}{2}|\mathbf{1}\rangle$$

The numbers θ and ϕ define a point on a 3-dimensional sphere known as the 'Bloch sphere'. Where 'α' and 'β' are complex components to the computation basis vectors, '$|0\rangle$' and '$|1\rangle$'; the quantum bits analogous to classical bits '0' and '1' respectively. $|\alpha|^2$ and $|\beta|^2$ give the probabilities of finding '$|0\rangle$' and '$|1\rangle$' respectively.

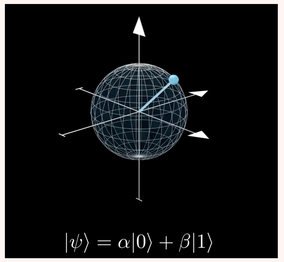


Figure 3. $|\psi\rangle$ in a *'Bloch Sphere'. Made using [12][9]*

A quantum computer encodes information using the quantum state of a composite system. It is to be emphasized that, when measured the qubit returns the computational basis vectors. Since the state of a qubit is altered the moment it is measured, it is not possible to utilize all possible values of α and β available on the bloch sphere. [7] [8] [9] [10]

Analogous to the way a classical computer is built from an electrical circuits of wires and logic gates, a quantum computer is also built from a quantum circuit containing wires and elementary quantum gates to carry around and manipulate quantum information. [8]

Classical Logic Gates

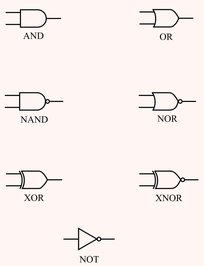


Figure 4. Classical Logic Gates.[6]

Quantum Gates

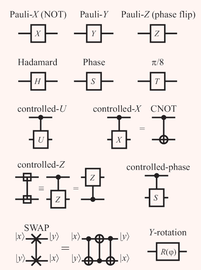


Figure 5. Quantum Gates. [10]

The matrix representation of the quantum gates is derived from the implied linearity property of quantum mechanics. The gates are essentially 2-by-2 matrices that can act on single or two-qubit systems to achieve a desired probability distribution in the qubits. For example, the classical NOT gate acts on a single-bit input and outputs an inverted bit. i.e., $NOT(A) = \overline{A}$. [8] [10]

In the quantum analog of the classical NOT gate, known as the Pauli-X (NOT) Gate, the complex vector components of the single qubit input are simply swapped.

Here, $X \equiv \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$; And consequently, application on a single qubit is written as, $X\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \begin{bmatrix}\beta\\ \alpha\end{bmatrix}$

The only inherent constraint on any 2-by-2 matrices (say, U) that may be used as a quantum gate is that they have to be unitary, i.e. $U^{\dagger}.U = I$; which essentially implies that the length and angles of the original complex vector in the bloch sphere remains unchanged and the total probability, $|\alpha|^2 + |\beta|^2 = 1$.

The most prominently utilized quantum gate is the Hadamard gate, denoted as, $H \equiv \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$; which sets α & β such that the probability of measuring $|\psi\rangle$ is 50% time $|0\rangle$ and 50% times $|1\rangle$. [8]

Function Implementation

A N-bit quantum register contains N qubits, formed using $n = 2^N$ computational basis states. For example, a 4-qubit register would have the basis states $|0000\rangle$ upto $|1111\rangle$. Similar to a 1-qubit system, the register also has all superposition states within $|\psi\rangle$. [10]

Inputs			Outputs		
a	b	c	a'	b'	c'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

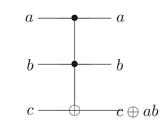


Figure 6. Truth table for a toffoli gate and its' circuit representation. [8]

In theory, just as functional logic circuits may be constructed from classical gates and registers; it is possible to construct quantum functional logic circuits using $q.gates$ and $q.registers$. [10] The quantum Toffoli gate's implementation ensures that quantum computers are in fact capable of performing any classical computation. [8]

Novel Algorithms

The existence of the quantum properties like $Parallelism$,$Entanglement$, $Teleportation$ and $Interference$ allows quantum computers to step beyond classical computers. They can be made to implement algorithms impossible for a classical counterpart. [8]

Groundbreaking Algorithms so far:

1. **Deutsch's Algorithm:** Determines if a function is constant or balanced using $parallelism$ and $interference$. [3]
2. **Shor's Algorithm:** Efficiently factors large integers and computes discrete logarithims using $fourier\ transform$ and $entanglement$.[11]
3. **Grover's Algorithm:** Searches an unsorted list faster than a classical computer using $parallelism$ and $interference$. [5]

Limitations & Conclusion

Qubit Decoherence, Error Correction, Control, Benchmarking and Scalability are major blockers to the quantum computing. With further development, there are major implications in the fields of cryptography, encryption and physical simulation fields. [2]

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